

Guiding THP Decisions with DAMON Memory Monitoring

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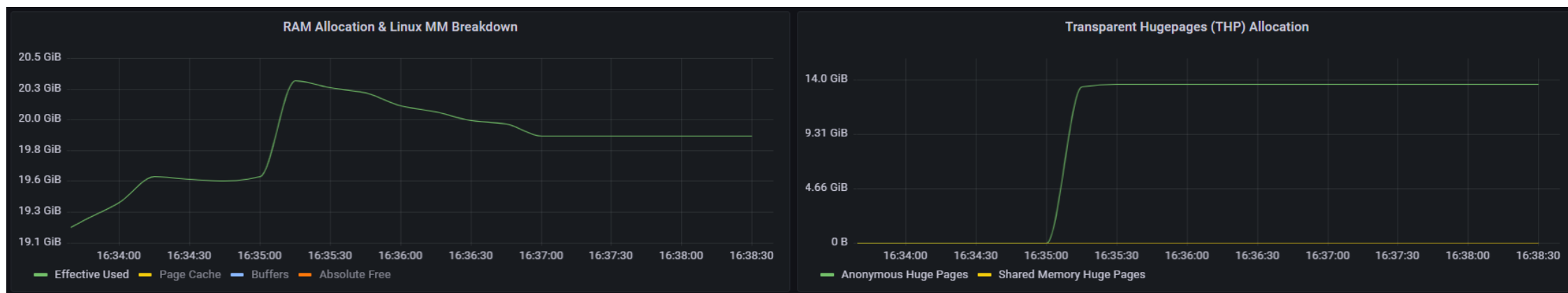


Motivations and goals

- Problems
 - > THP is global and affects the entire system
 - > Huge pages can lead to fragmentation, memory waste and OOM
- Goals
 - > THP automated management
 - > Custom THP policies per task instead of global THP policy
 - > Bypass khugepaged

MADV_COLLAPSE in DAMON

- Action “collapse” was introduced, supporting collapsing pages into huge pages
- Collapse may work for specific memory regions triggered by DAMON, as opposed to the entire VMA
- It works with THP set to never. It doesn't require khugepaged to be running
- It complements the current hugepage action, which internally uses madvise



Autotuning huge pages

- hugepage_mem_bp target metric for quota tuning introduced
- Huge page to total memory used ratio in basis points (1/10,000)
- It uses anonymous + shared memory + file mapped sizes

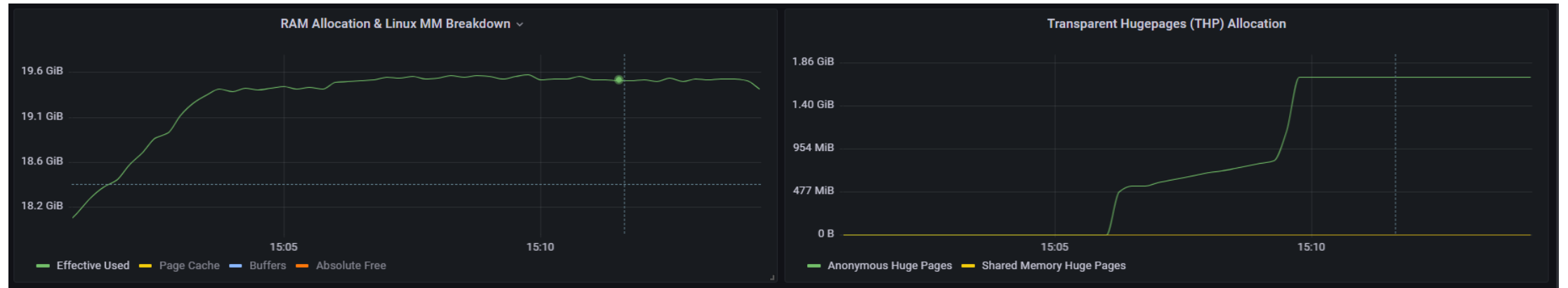
Testing

- Same setting as before

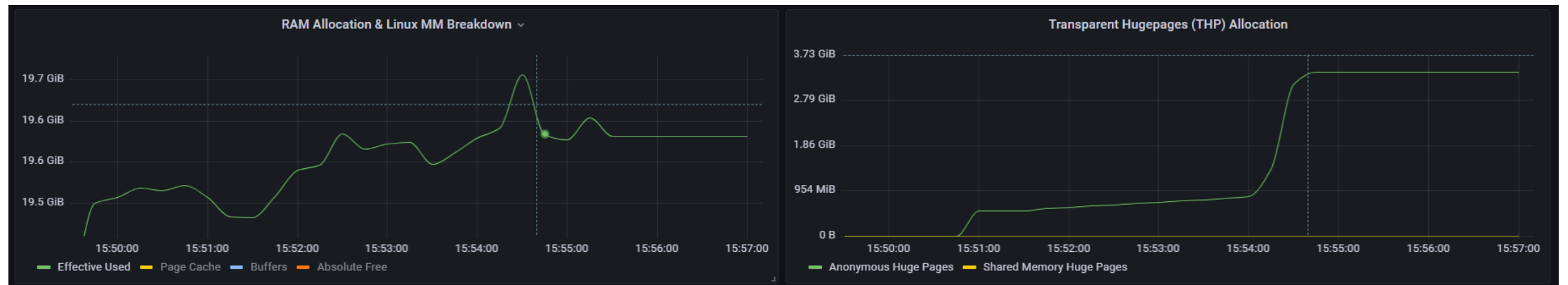
```
sudo ./damo start $(pidof mariadb) --monitoring_nr_regions_range 10 1000 --monitoring_intervals 5000 100000 600000000 \
--damos_quota_time 0 --damos_quota_space 128000000 --damos_quota_interval 1000 --damos_quota_weights 0 1 1 \
--damos_quota_goal hugepage_mem_bp 1000 --damos_quota_goal_tuner temporal --damos_apply_interval 100000 \
--damos_access_rate 0 max --damos_age 50 max --damos_action collapse --debug_damon
```

Results

Temporal autotuning



Consistent autotuning



Future directions

- mTHP support for hugepage_mem_bp
- Split support for full THP management using DAMON